

Initiating Search

November 10, 2025, 4:06 PM

• Search:

CAS Lexicon Search:

Fluorination, electrophilic - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (455)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (285)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

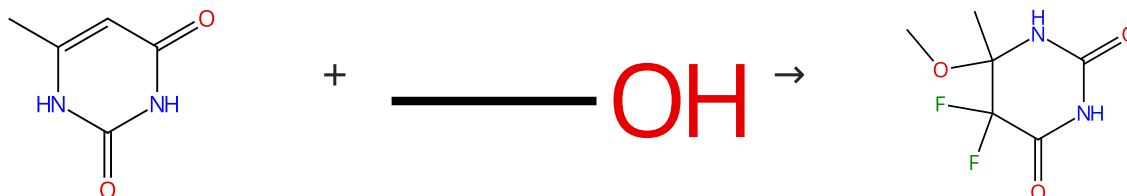
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (88) Suppliers (444)

31-119-CAS-15820944

Steps: 1 Yield: 100%

Promotional effect of ionic liquids in electro philic fluorination of methylated uracils

By: Borodkin, Gennady I.; et al

RSC Advances (2016), 6(65), 60556-60564.

1.1 Reagents: Selectfluor

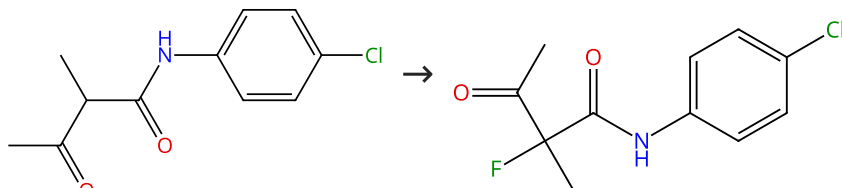
Catalysts: 1-Butyl-3-methylimidazolium trifluoromethanesulfonate

Solvents: Methanol; 4 h, 80 °C

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (3) Supplier (1)

31-119-CAS-3871683

Steps: 1 Yield: 100%

Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400

By: Bi, Jingjing; et al

Green Chemistry (2012), 14(4), 1159-1162.

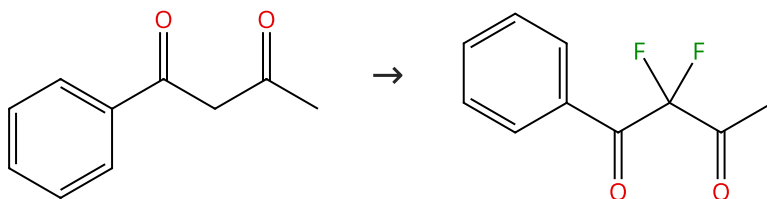
1.1 Reagents: Selectfluor

Solvents: Polyethylene glycol; 14 h, 60 °C

Experimental Protocols

Scheme 3 (1 Reaction)

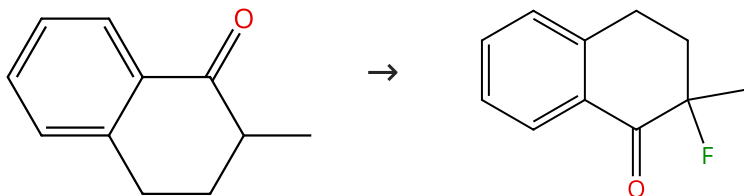
Steps: 1 Yield: 100%

 Suppliers (81) Suppliers (21)

31-119-CAS-6566125 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine trihydrofluoride, 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole Solvents: Dichloromethane; 24 h, 60 °C Experimental Protocols	Electrophilic fluorination using a hypervalent iodine reagent derived from fluoride By: Geary, Gemma C.; et al Chemical Communications (Cambridge, United Kingdom) (2013), 49(81), 9263-9265.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



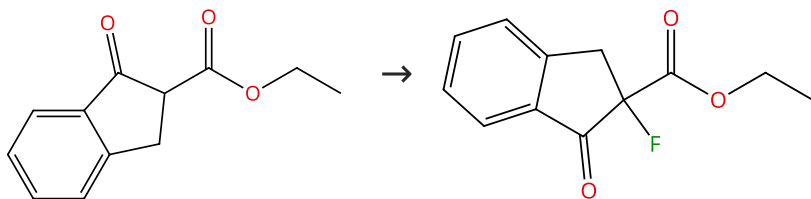
Suppliers (65)

Suppliers (8)

31-119-CAS-1523793 Steps: 1 Yield: 100% 1.1 Reagents: Sodium bis(trimethylsilyl)amide Solvents: Diethyl ether 1.2 Reagents: 1,3,2-Benzodithiazole, 2-fluoro-, 1,1,3,3-tetraoxide Solvents: Diethyl ether	N-Fluoro-o-benzenedisulfonimide: a useful new fluorinating reagent By: Davis, Franklin A.; et al Tetrahedron Letters (1991), 32(13), 1631-4.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



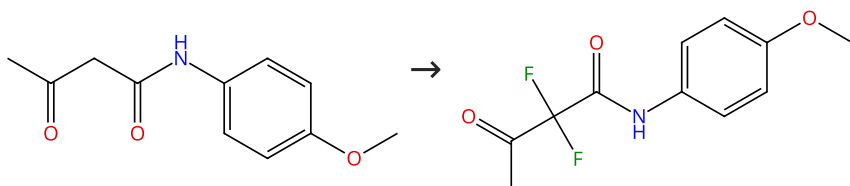
Suppliers (48)

Supplier (1)

31-119-CAS-10809458 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine trihydrofluoride, 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole; 48 h, 60 °C Experimental Protocols	Electrophilic fluorination using a hypervalent iodine reagent derived from fluoride By: Geary, Gemma C.; et al Chemical Communications (Cambridge, United Kingdom) (2013), 49(81), 9263-9265.
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



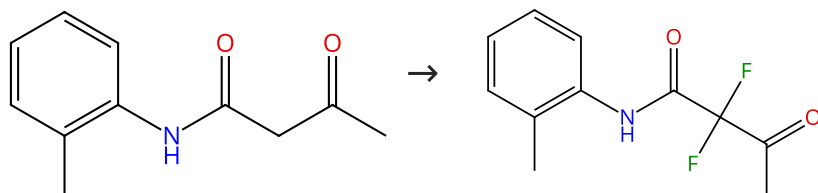
Suppliers (75)

Suppliers (2)

31-119-CAS-15171701	Steps: 1 Yield: 99%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 16 h, rt		

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%



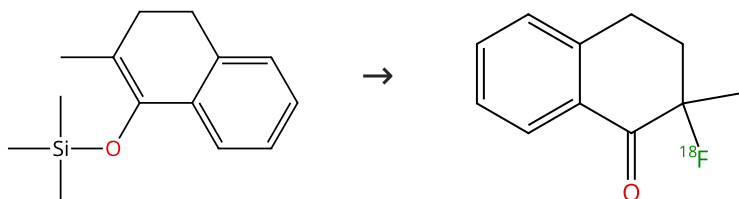
Suppliers (72)

Suppliers (2)

31-119-CAS-6647636	Steps: 1 Yield: 99%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 8 h, rt		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%

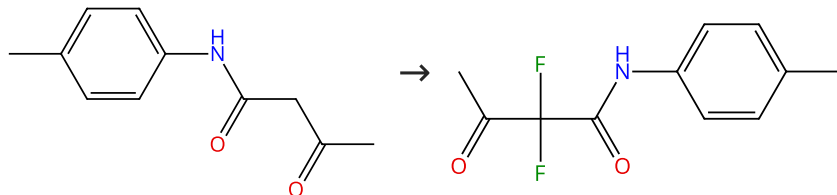


Suppliers (3)

31-119-CAS-21164207	Steps: 1 Yield: 99%	Synthesis and reactivity of [^{18}F]-N-fluorobenzenesulfonimide By: Teare, Harriet; et al Chemical Communications (Cambridge, United Kingdom) (2007), (23), 2330-2332.
1.1 Reagents: <i>N</i> -(Fluoro- ^{18}F)- <i>N</i> -(phenylsulfonyl)benzenesulfonamide Solvents: Acetonitrile; 15 min, rt		
Experimental Protocols		

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



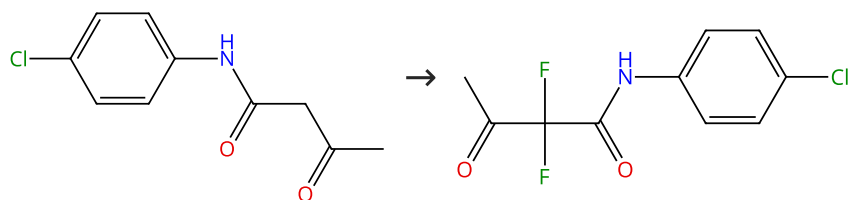
Suppliers (50)

Suppliers (2)

31-119-CAS-852598	Steps: 1 Yield: 99%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 2 h, rt		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (63)

Suppliers (2)

31-119-CAS-4187060

Steps: 1 Yield: 99%

The synthesis of α,α -difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400

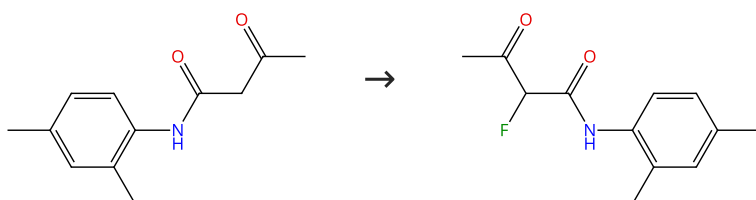
By: Zhang, Zhiguo; et al

Journal of Fluorine Chemistry (2013), 151, 45-49.

- 1.1 Reagents: Potassium carbonate
Solvents: Water, Polyethylene glycol; 1 h, rt
- 1.2 Reagents: Selectfluor; 8 h, rt

Scheme 11 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (67)

Supplier (1)

31-119-CAS-15244243

Steps: 1 Yield: 99%

Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400

By: Bi, Jingjing; et al

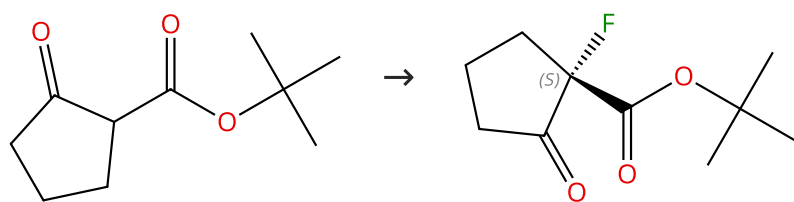
Green Chemistry (2012), 14(4), 1159-1162.

- 1.1 Reagents: Selectfluor
Solvents: Polyethylene glycol; 8 h, 60 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (44)

Absolute stereochemistry shown,
Rotation (-)

31-119-CAS-11551008

Steps: 1 Yield: 99%

Asymmetric fluorination of β -keto esters catalyzed by chiral rare earth perfluorinated organophosphates

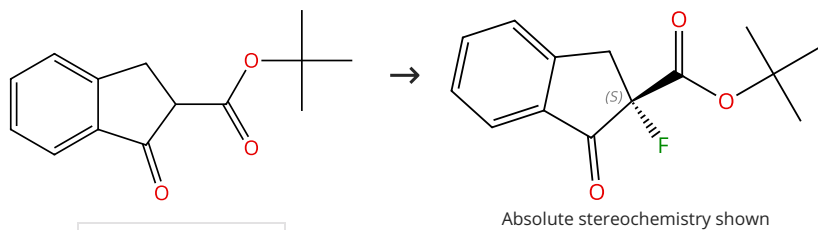
By: Suzuki, Shoko; et al

Tetrahedron: Asymmetry (2006), 17(4), 504-507.

- 1.1 Reagents: Pyridinium, 1-fluoro-, 1,1,1-trifluoromethanesulfonate (1:1)
Catalysts: Scandium, tris[(11b*R*)-8,9,10,11,12,13,14,15-octafuoro-4-(hydroxy- κO)dinaphtho[2,1-*d'*:1',2'-*f*][1,3,2]dioxaphosphen 4-(oxidato- κO)]-, (OC-6-11)-
Solvents: Toluene; 6 h, rt

Scheme 13 (2 Reactions)

Steps: 1 Yield: 96-99%



Suppliers (5)

31-119-CAS-3071819

Steps: 1 Yield: 99%

Designing Fluorinated Cinchona Alkaloids for Enantioselective Catalysis: Controlling Internal Rotation by a Fluorine-Ammonium Ion gauche Effect (φ_{NCCF})

By: Tanzer, Eva-Maria; et al

Chemistry - A European Journal (2012), 18(7), 2006-2013, S2006/1-S2006/41.

1.1 **Reagents:** Cesium carbonate
Catalysts: Cinchonanium, 9-fluoro-6'-hydroxy-1-(phenylmethyl)-, bromide (1:1), (9*S*)-
Solvents: Toluene; 1.5 h, rt

1.2 **Reagents:** *N*-Fluorobenzenesulfonimide; rt

1.3 **Reagents:** Water; rt

Experimental Protocols

31-119-CAS-14403678

Steps: 1 Yield: 96%

Designing Fluorinated Cinchona Alkaloids for Enantioselective Catalysis: Controlling Internal Rotation by a Fluorine-Ammonium Ion gauche Effect (φ_{NCCF})

By: Tanzer, Eva-Maria; et al

Chemistry - A European Journal (2012), 18(7), 2006-2013, S2006/1-S2006/41.

1.1 **Reagents:** Cesium carbonate
Catalysts: Cinchonanium, 9-fluoro-1-(phenylmethyl)-, chloride (1:1), (9*S*)-
Solvents: Toluene; 1.5 h, rt

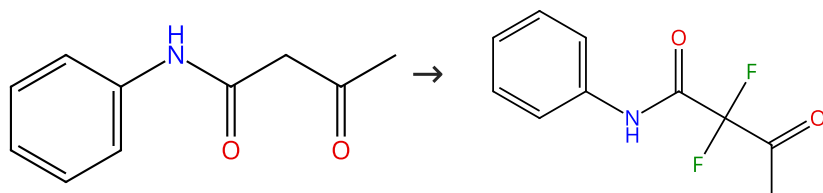
1.2 **Reagents:** *N*-Fluorobenzenesulfonimide; 2 h, rt

1.3 **Reagents:** Water; rt

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (59)

Suppliers (2)

31-119-CAS-15457668

Steps: 1 Yield: 99%

The synthesis of α,α -difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400

By: Zhang, Zhiguo; et al

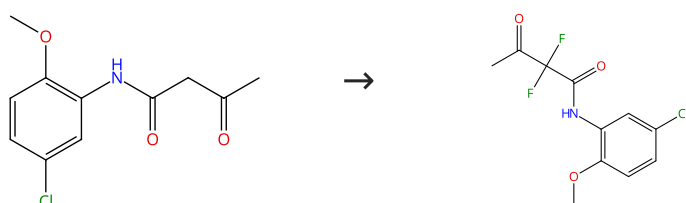
Journal of Fluorine Chemistry (2013), 151, 45-49.

1.1 **Reagents:** Potassium carbonate
Solvents: Water, Polyethylene glycol; 1 h, rt

1.2 **Reagents:** Selectfluor; 5 h, rt

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



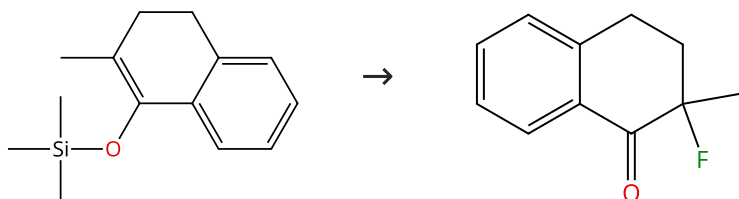
Suppliers (39)

Suppliers (2)

31-119-CAS-8452338	Steps: 1 Yield: 98%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 10 h, rt		

Scheme 16 (1 Reaction)

Steps: 1 Yield: 98%



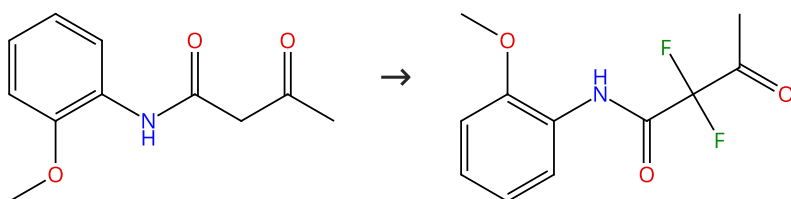
Suppliers (3)

Suppliers (8)

31-119-CAS-21382696	Steps: 1 Yield: 98%	Radiosynthesis and Evaluation of [^{18}F]Selectfluor bis(triflate) By: Teare, Harriet; et al Angewandte Chemie, International Edition (2010), 49(38), 6821-6824, S6821/1-S6821/19.
1.1 Reagents: Selectfluor Solvents: Acetonitrile; 1.5 h, rt		
1.2 Reagents: Water; rt Experimental Protocols		

Scheme 17 (1 Reaction)

Steps: 1 Yield: 98%



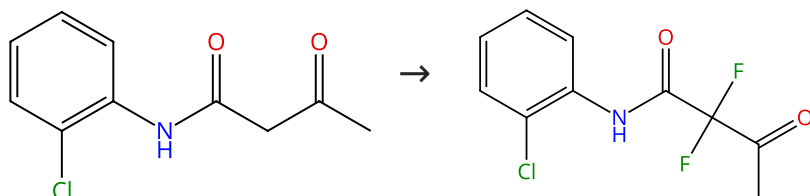
Suppliers (68)

Suppliers (2)

31-119-CAS-10913334	Steps: 1 Yield: 98%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 4 h, rt		

Scheme 18 (1 Reaction)

Steps: 1 Yield: 98%



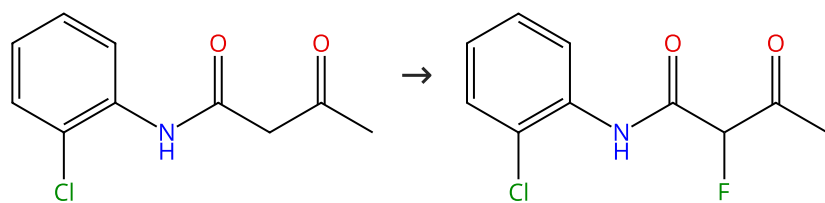
Suppliers (40)

Suppliers (2)

31-119-CAS-2182132	Steps: 1 Yield: 98%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 5 h, rt		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (40)

Suppliers (2)

31-119-CAS-4258598

Steps: 1 Yield: 98%

Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400

1.1 Reagents: Selectfluor
Solvents: Polyethylene glycol; 8 h, 80 °C

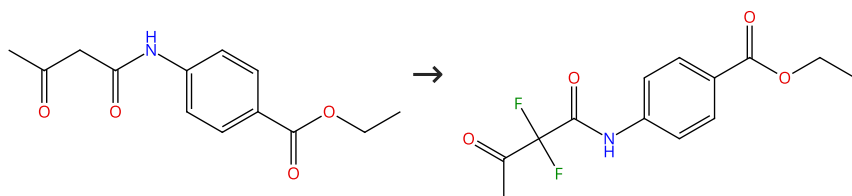
Experimental Protocols

By: Bi, Jingjing; et al

Green Chemistry (2012), 14(4), 1159-1162.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (43)

Supplier (1)

31-119-CAS-12706915

Steps: 1 Yield: 98%

The synthesis of α,α -difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400

1.1 Reagents: Potassium carbonate
Solvents: Water, Polyethylene glycol; 1 h, rt

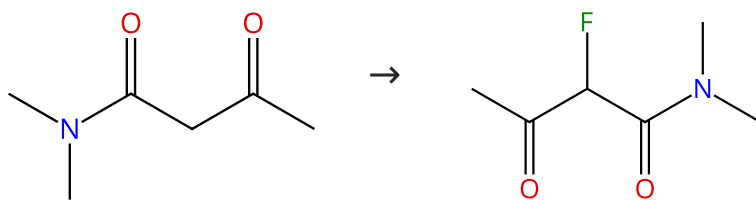
1.2 Reagents: Selectfluor; 7 h, rt

By: Zhang, Zhiguo; et al

Journal of Fluorine Chemistry (2013), 151, 45-49.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (57)

Suppliers (6)

31-119-CAS-10326893

Steps: 1 Yield: 98%

Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400

1.1 Reagents: Selectfluor
Solvents: Polyethylene glycol; 11 h, 60 °C

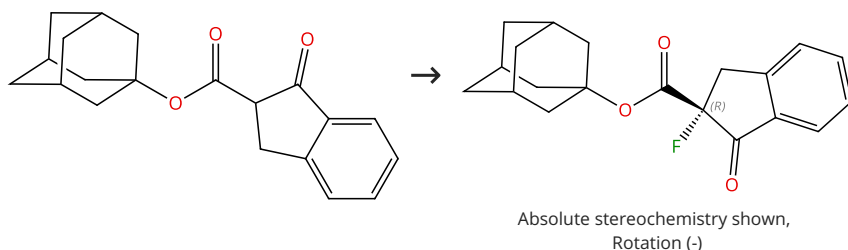
Experimental Protocols

By: Bi, Jingjing; et al

Green Chemistry (2012), 14(4), 1159-1162.

Scheme 22 (2 Reactions)

Steps: 1 Yield: 95-98%



31-119-CAS-11285247

Steps: 1 Yield: 98%

Designing Fluorinated Cinchona Alkaloids for Enantioselective Catalysis: Controlling Internal Rotation by a Fluorine-Ammonium Ion gauche Effect (φ_{NCCF})

By: Tanzer, Eva-Maria; et al

Chemistry - A European Journal (2012), 18(7), 2006-2013, S2006/1-S2006/41.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Cinchonanium, 9-fluoro-6'-methoxy-1-(phenyl methyl)-, chloride (1:1), (9S)-
Solvents: Toluene; 1.5 h, rt
- 1.2 **Reagents:** *N*-Fluorobenzenesulfonimide; rt
- 1.3 **Reagents:** Water; rt

Experimental Protocols

31-119-CAS-20069649

Steps: 1 Yield: 95%

Catalytic asymmetric fluorination of alkyl 1-indanone-2-carboxylates ruled by pybox-Eu(III) combination

By: Granados, Albert; et al

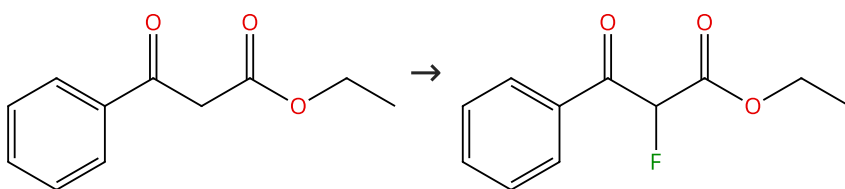
Molecules (2019), 24(6), 1141/1-1141/10.

- 1.1 **Catalysts:** Lanthanum triflate, (3a*S*,3'a*S*,8a*R*,8'a*R*)-2,2'-(2,6-Pyridinediyl)bis[3a,8a-dihydro-8*H*-indeno[1,2-*d*]oxazole]
Solvents: Acetonitrile; overnight, rt
- 1.2 1 h, rt; rt → -30 °C
- 1.3 **Reagents:** *N*-Fluorobenzenesulfonimide; -30 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

Suppliers (37)

31-119-CAS-11509245

Steps: 1 Yield: 98%

Multiple Microcapillary Reactor for Organic Synthesis

By: Hornung, Christian H.; et al

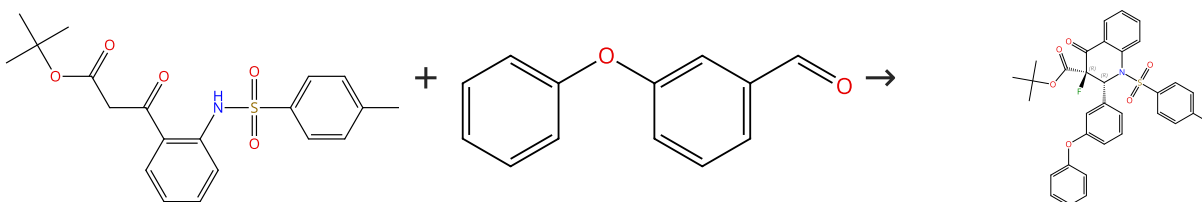
Industrial & Engineering Chemistry Research (2010), 49(10), 4576-4582.

- 1.1 **Reagents:** Selectfluor
Solvents: Acetonitrile; 0.5 h, 120 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%



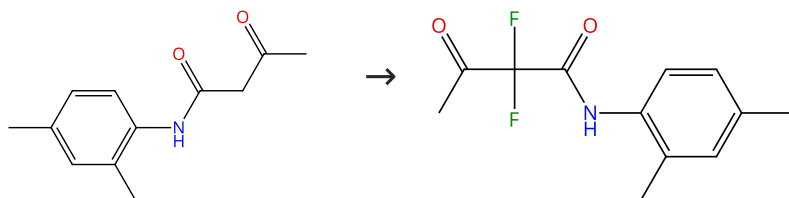
Relative stereochemistry shown

Suppliers (85)

31-119-CAS-6061851	Steps: 1 Yield: 98%	Stereoselective synthesis of fluorinated 2,3-dihydroquinolin-4 (1H)-ones via a one-pot multistep transformation
1.1 Catalysts: Pyrrolidine Solvents: Ethanol; 24 h, rt		By: Li, Feng; et al
1.2 Reagents: Sodium carbonate, <i>N</i> -Fluorobenzenesulfonimide		Journal of Organic Chemistry (2012), 77(5), 2398-2406.
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



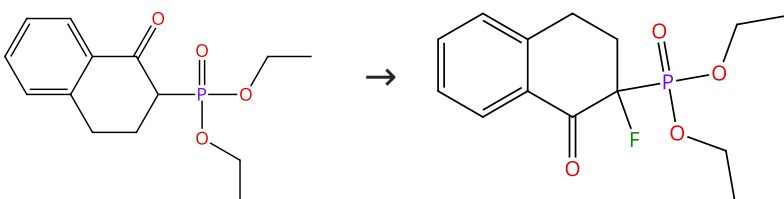
Suppliers (67)

Suppliers (2)

31-119-CAS-8784664	Steps: 1 Yield: 98%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		By: Zhang, Zhiguo; et al
1.2 Reagents: Selectfluor; 10 h, rt		Journal of Fluorine Chemistry (2013), 151, 45-49.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (3)

31-119-CAS-9976940	Steps: 1 Yield: 97%	An efficient catalytic enantioselective fluorination of β-ketophosphonates using chiral palladium complexes
1.1 Reagents: <i>N</i> -Fluorobenzenesulfonimide Catalysts: Palladium(2+), diaqua[1,1'-(1 <i>R</i>)-[1,1'-binaphthalene]-2,2'-diylbis[1,1-bis(3,5-dimethylphenyl)phosphine- κ P]]-, (<i>SP</i> -4-2)-, 1,1,1-trifluoromethanesulfonate (1:2) Solvents: Ethanol; 3 h, rt		By: Hamashima, Yoshitaka; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Tetrahedron Letters (2005), 46(9), 1447-1450.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 97%

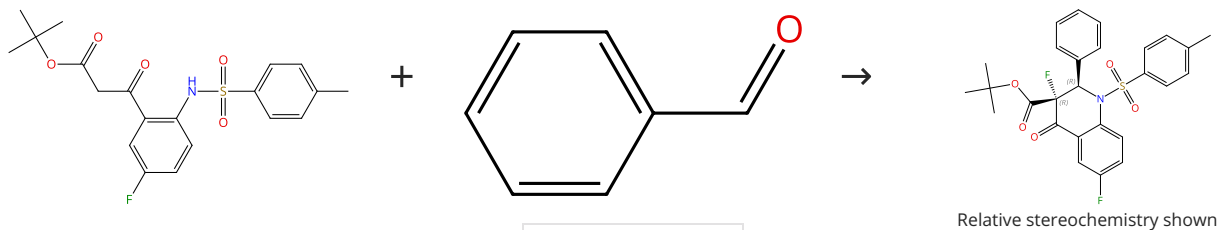


Suppliers (64)

31-119-CAS-13157498	Steps: 1 Yield: 97%	Electrophilic fluorination of 6-methyl- and 1,3,6-trimethyluracils in water
1.1 Reagents: Selectfluor Solvents: Water; 4 h, 80 °C		By: Borodkin, G. I.; et al
Experimental Protocols		Russian Journal of Organic Chemistry (2015), 51(7), 1003-1007.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 97%

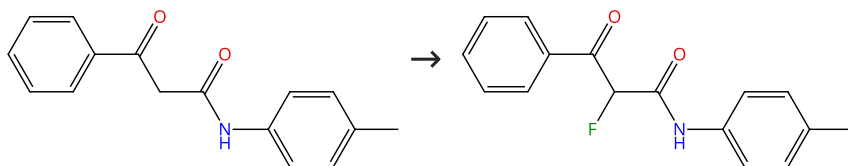


Suppliers (69)

31-119-CAS-13880391	Steps: 1 Yield: 97%	Stereoselective synthesis of fluorinated 2,3-dihydroquinolin-4(1H)-ones via a one-pot multistep transformation
1.1 Catalysts: Pyrrolidine Solvents: Ethanol; 24 h, rt		By: Li, Feng; et al
1.2 Reagents: Sodium carbonate, <i>N</i> -Fluorobenzenesulfonimide		Journal of Organic Chemistry (2012), 77(5), 2398-2406.
Experimental Protocols		

Scheme 29 (1 Reaction)

Steps: 1 Yield: 97%



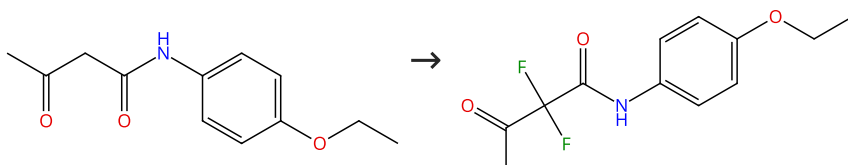
Suppliers (12)

Supplier (1)

31-119-CAS-7865812	Steps: 1 Yield: 97%	Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400
1.1 Reagents: Selectfluor Solvents: Polyethylene glycol; 2 h, 60 °C		By: Bi, Jingjing; et al
Experimental Protocols		Green Chemistry (2012), 14(4), 1159-1162.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 97%



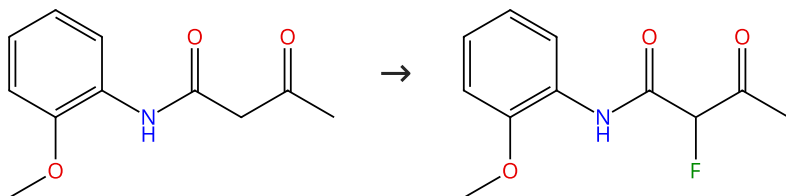
Suppliers (57)

Suppliers (2)

31-119-CAS-3909889	Steps: 1 Yield: 97%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt		
1.2 Reagents: Selectfluor; 11 h, rt		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 97%



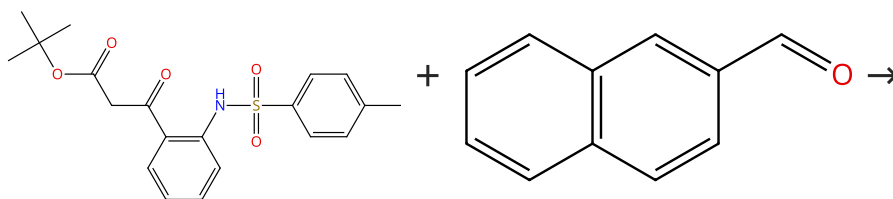
Suppliers (68)

Suppliers (2)

31-119-CAS-13115314	Steps: 1 Yield: 97%	Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400 By: Bi, Jingjing; et al Green Chemistry (2012), 14(4), 1159-1162.
1.1 Reagents: Selectfluor Solvents: Polyethylene glycol; 14 h, 60 °C		
Experimental Protocols		

Scheme 32 (1 Reaction)

Steps: 1 Yield: 97%



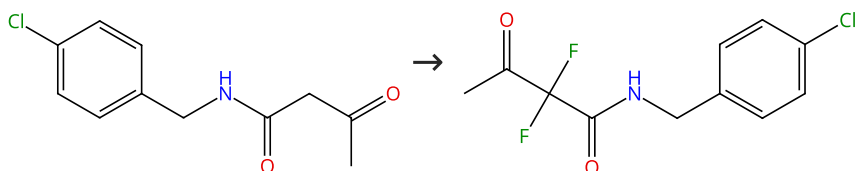
Relative stereochemistry shown

Suppliers (86)

31-119-CAS-5680041	Steps: 1 Yield: 97%	Stereoselective synthesis of fluorinated 2,3-dihydroquinolin-4(1H)-ones via a one-pot multistep transformation By: Li, Feng; et al Journal of Organic Chemistry (2012), 77(5), 2398-2406.
1.1 Catalysts: Pyrrolidine Solvents: Ethanol; 24 h, rt		
1.2 Reagents: Sodium carbonate, <i>N</i> -Fluorobenzenesulfonimide Experimental Protocols		

Scheme 33 (1 Reaction)

Steps: 1 Yield: 96%



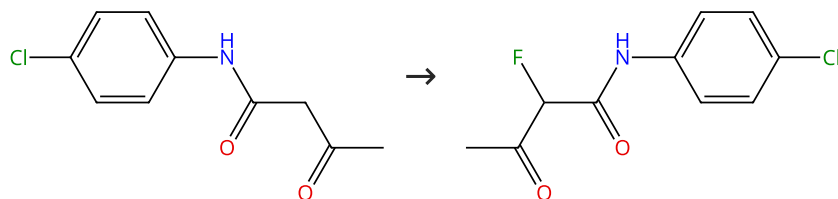
Suppliers (34)

Suppliers (2)

31-119-CAS-1743544 Steps: 1 Yield: 96%	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt	
1.2 Reagents: Selectfluor; 10 h, rt	

Scheme 34 (1 Reaction)

Steps: 1 Yield: 96%



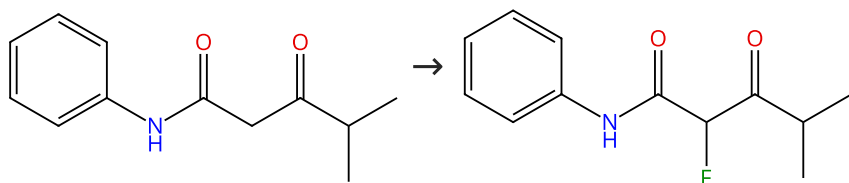
Suppliers (63)

Suppliers (2)

31-119-CAS-8518232 Steps: 1 Yield: 96%	Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400 By: Bi, Jingjing; et al Green Chemistry (2012), 14(4), 1159-1162.
1.1 Reagents: Selectfluor Solvents: Polyethylene glycol; 5 h, 60 °C	
Experimental Protocols	

Scheme 35 (1 Reaction)

Steps: 1 Yield: 96%



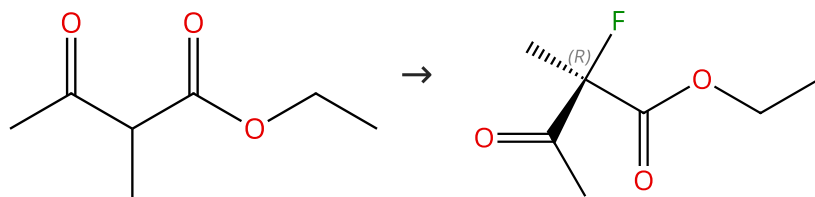
Suppliers (66)

Suppliers (2)

31-119-CAS-3604809 Steps: 1 Yield: 96%	Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400 By: Bi, Jingjing; et al Green Chemistry (2012), 14(4), 1159-1162.
1.1 Reagents: Selectfluor Solvents: Polyethylene glycol; 17 h, 60 °C	
Experimental Protocols	

Scheme 36 (1 Reaction)

Steps: 1 Yield: 96%


 Absolute stereochemistry shown,
 Rotation (-)

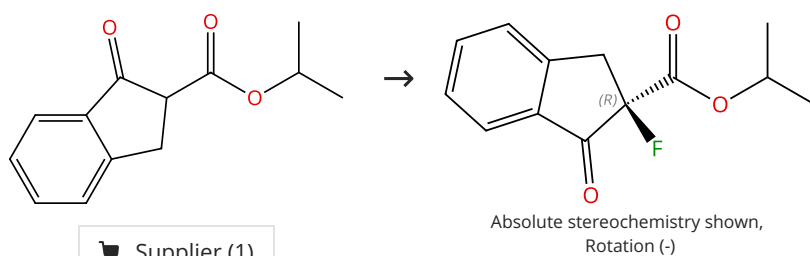
Suppliers (86)

Supplier (1)

31-119-CAS-3547021 Steps: 1 Yield: 96% 1.1 Catalysts: Oxonium, triethyl-, hexafluorophosphate(1-) (1:1), (<i>OC</i>-6-13)-[(1<i>S</i>,2<i>S</i>)-<i>N</i>¹,<i>N</i>²-Bis[[2-(diphenylphosphino-<i>κP</i>)phenyl]methylene]-1,2-cyclohexanediamine-<i>κN</i>¹,<i>κN</i>²]dichlororuthenium Solvents: Dichloromethane; overnight, rt 1.2 Solvents: Diethyl ether; 10 min, rt 1.3 Reagents: <i>N</i>-Fluorobenzenesulfonimide; 4 h, rt 1.4 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Ruthenium-Catalyzed Asymmetric Electrophilic Fluorination of 1,3-Dicarbonyl Compounds By: Althaus, Martin; et al Organometallics (2007), 26(24), 5902-5911.
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Scheme 37 (1 Reaction)

Steps: 1 Yield: 96%

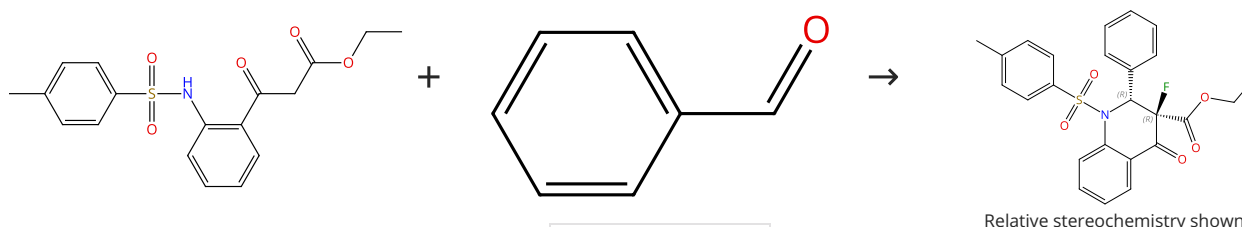


Supplier (1)

31-119-CAS-9548287 Steps: 1 Yield: 96% 1.1 Reagents: 4-Morpholineethanesulfonic acid, Selectfluor Catalysts: Copper, (4,4'-dimethyl-2,2'-bipyridine-<i>κN</i>¹,<i>κN</i>^{1'})bis(nitrato-<i>κO</i>)-, (<i>SP</i>-4-2)- Solvents: Acetonitrile, Water; 6 h, pH 5.5, 5 °C Experimental Protocols	DNA-mediated enantioselective carbon-fluorine bond formation By: Shibata, Norio; et al Synlett (2007), (7), 1153-1157.
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Scheme 38 (1 Reaction)

Steps: 1 Yield: 95%

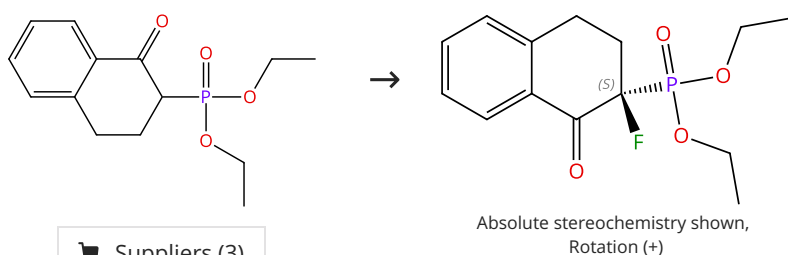


Suppliers (69)

31-119-CAS-5343567 Steps: 1 Yield: 95% 1.1 Catalysts: Pyrrolidine Solvents: Ethanol; 24 h, rt 1.2 Reagents: Sodium carbonate, <i>N</i>-Fluorobenzenesulfonimide Experimental Protocols	Stereoselective synthesis of fluorinated 2,3-dihydroquinolin-4(1H)-ones via a one-pot multistep transformation By: Li, Feng; et al Journal of Organic Chemistry (2012), 77(5), 2398-2406.
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Scheme 39 (1 Reaction)

Steps: 1 Yield: 95%

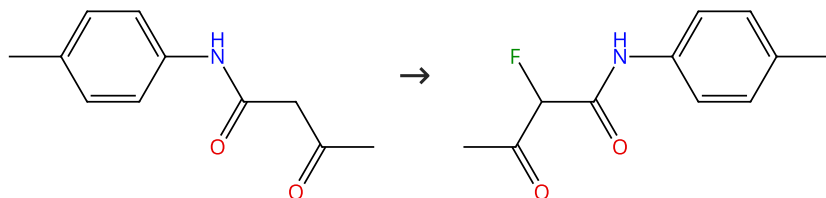


Suppliers (3)

31-119-CAS-11541229 Steps: 1 Yield: 95% 1.1 Reagents: <i>N</i>-Fluorobenzenesulfonimide Catalysts: Palladium(2+), (acetonitrile)aqua[1,1'-(1<i>R</i>)-[1,1'-binaphthalene]-2,2'-diylbis[1,1-diphenylphosphine-κ<i>P</i>]]-, (<i>SP</i>-4-3)-, tetrafluoroborate(1-) (1:2) Solvents: 1-Butyl-3-methylimidazolium tetrafluoroborate; 10 h, rt 1.2 Reagents: Diethyl ether; rt	Asymmetric electrophilic fluorination of β-keto phosphonates in ionic liquid media catalyzed by chiral palladium complexes By: Kim, Sung Min; et al Bulletin of the Korean Chemical Society (2006), 27(3), 423-425.
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Scheme 40 (1 Reaction)

Steps: 1 Yield: 95%



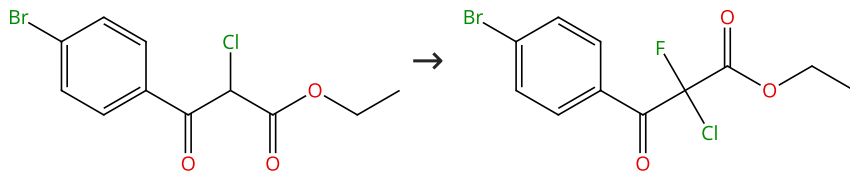
Suppliers (50)

Suppliers (2)

31-119-CAS-13675526 Steps: 1 Yield: 95% 1.1 Reagents: Selectfluor Solvents: Polyethylene glycol; 8 h, 60 °C Experimental Protocols	Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400 By: Bi, Jingjing; et al Green Chemistry (2012), 14(4), 1159-1162.
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Scheme 41 (1 Reaction)

Steps: 1 Yield: 94%

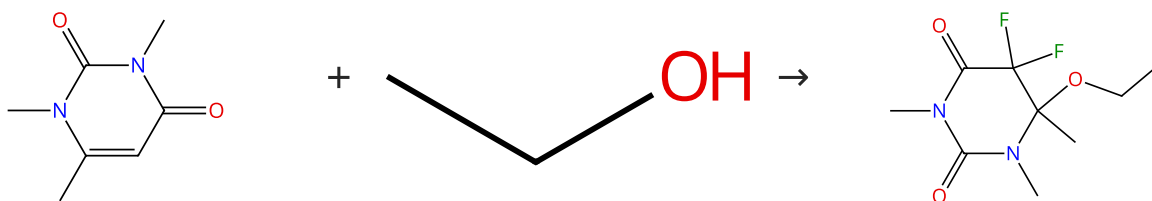


Suppliers (5)

31-119-CAS-4246192 Steps: 1 Yield: 94% 1.1 Reagents: <i>N</i>-Fluorobenzenesulfonimide Catalysts: Palladium(2+), (acetonitrile)aqua[1,1'-(1<i>R</i>)-[1,1'-binaphthalene]-2,2'-diylbis[1,1-diphenylphosphine-κ<i>P</i>]]-, (<i>SP</i>-4-3)-, hexafluorophosphate(1-) (1:2) Solvents: Methanol; 72 h, rt	Catalytic asymmetric fluorination of α-chloro-β-ketoesters in the presence of chiral palladium complexes By: Cho, Min Je; et al Bulletin of the Korean Chemical Society (2007), 28(12), 2191-2192.
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Scheme 42 (1 Reaction)

Steps: 1 Yield: 94%



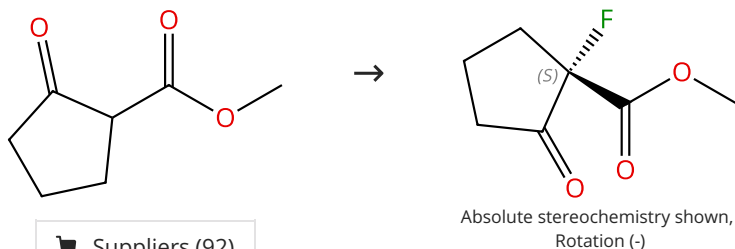
Suppliers (64)

Suppliers (620)

31-119-CAS-15923569 Steps: 1 Yield: 94% 1.1 Reagents: Selectfluor Catalysts: 1-Butyl-3-methylimidazolium trifluoromethanesulfonate Solvents: Ethanol; 10 h, 80 °C Experimental Protocols	Promotional effect of ionic liquids in electrophilic fluorination of methylated uracils By: Borodkin, Gennady I.; et al RSC Advances (2016), 6(65), 60556-60564.
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Scheme 43 (1 Reaction)

Steps: 1 Yield: 94%



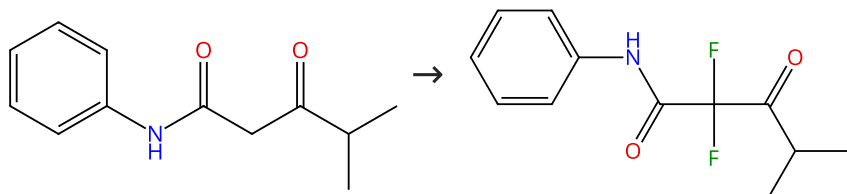
Suppliers (92)

Suppliers (3)

31-119-CAS-77033 Steps: 1 Yield: 94% 1.1 Reagents: Pyridinium, 1-fluoro-, 1,1,1-trifluoromethanesulfonate (1:1) Catalysts: Scandium, tris[(11b<i>R</i>)-8,9,10,11,12,13,14,15-octafuoro-4-(hydroxy-κO)dinaphtho[2,1-<i>d'</i>:1',2'-<i>f</i>][1,3,2]dioxaphosphepin 4-(oxidato-κO)]-, (OC-6-11)- Solvents: Toluene; 24 h, rt	Asymmetric fluorination of β-keto esters catalyzed by chiral rare earth perfluorinated organophosphates By: Suzuki, Shoko; et al Tetrahedron: Asymmetry (2006), 17(4), 504-507.
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Scheme 44 (1 Reaction)

Steps: 1 Yield: 94%



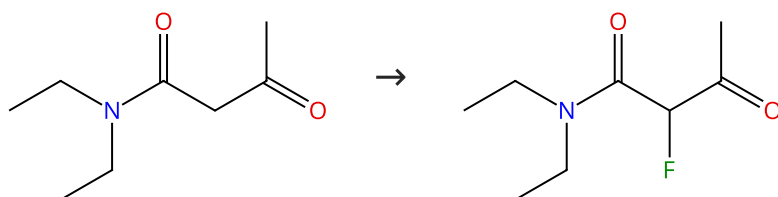
Suppliers (66)

Suppliers (2)

31-119-CAS-5991323 Steps: 1 Yield: 94% 1.1 Reagents: Potassium carbonate Solvents: Water, Polyethylene glycol; 1 h, rt 1.2 Reagents: Selectfluor; 15 h, rt	The synthesis of α,α-difluoroacetamides via electrophilic fluorination in the mixed-solvent of water and PEG-400 By: Zhang, Zhiguo; et al Journal of Fluorine Chemistry (2013), 151, 45-49.
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Scheme 45 (1 Reaction)

Steps: 1 Yield: 94%



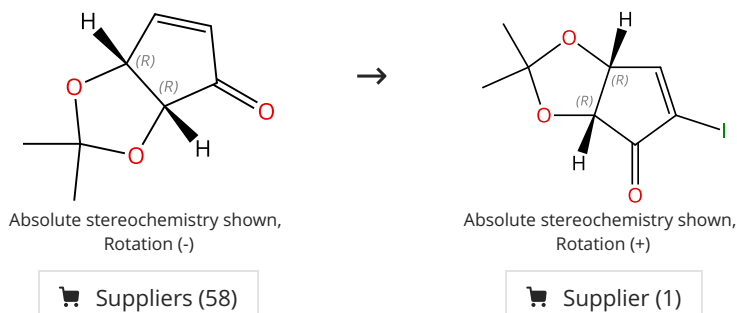
Suppliers (69)

Suppliers (15)

31-119-CAS-6344890	Steps: 1 Yield: 94%	Catalyst-free and highly selective electrophilic mono-fluorination of acetoacetamides: facile and efficient preparation of 2-fluoroacetoacetamides in PEG-400
1.1 Reagents: Selectfluor Solvents: Polyethylene glycol; 8 h, 60 °C		By: Bi, Jingjing; et al
Experimental Protocols		Green Chemistry (2012), 14(4), 1159-1162.

Scheme 46 (1 Reaction)

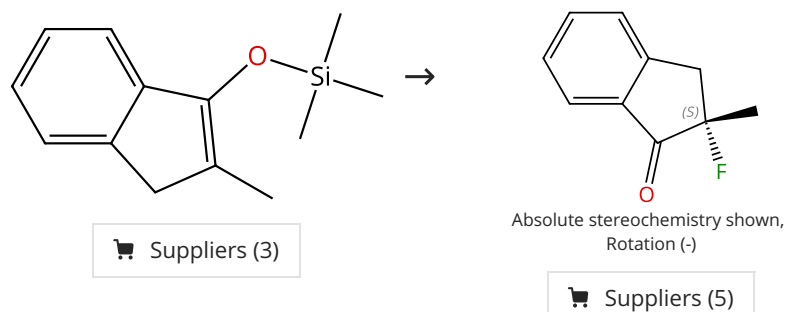
Steps: 1 Yield: 94%



31-119-CAS-13305349	Steps: 1 Yield: 94%	Synthesis of fluorinated cyclopentenyladenine as potent inhibitor of S-adenosylhomocysteine hydrolase
1.1 Reagents: Pyridine, Iodine Solvents: Dichloromethane; 1 h, rt		By: Kim, Hea Ok; et al
		Bioorganic & Medicinal Chemistry Letters (2004), 14(9), 2091-2093.

Scheme 47 (1 Reaction)

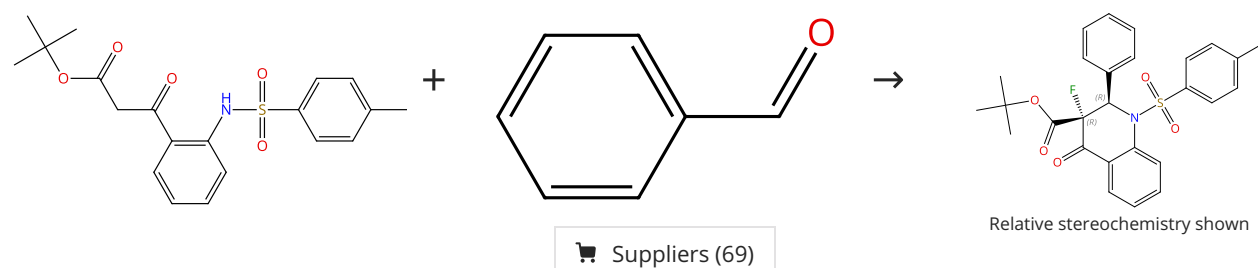
Steps: 1 Yield: 94%



31-119-CAS-21278012	Steps: 1 Yield: 94%	Enantioselective electrophilic fluorination: a study of the fluorine-transfer from achiral N-F reagents to cinchona alkaloids
1.1 Reagents: N-Fluorobenzenesulfonimide, Cinchonon-9-ol, 6'-methoxy-, 9-(4-chlorobenzoate), (8 <i>α</i> ,9 <i>R</i>)- Solvents: Acetonitrile; 1 h, rt; rt → -40 °C		By: Baudequin, Christine; et al
1.2 Solvents: Acetonitrile; 12 h, -40 °C		Journal of Fluorine Chemistry (2003), 122(2), 189-193.
1.3 Reagents: Water		

Scheme 48 (1 Reaction)

Steps: 1 Yield: 94%



31-119-CAS-9178470	Steps: 1 Yield: 94%	Stereoselective synthesis of fluorinated 2,3-dihydroquinolin-4 (1H)-ones via a one-pot multistep transformation By: Li, Feng; et al Journal of Organic Chemistry (2012), 77(5), 2398-2406.
1.1 Catalysts: Pyrrolidine Solvents: Ethanol; 24 h, rt		
1.2 Reagents: Sodium carbonate, <i>N</i> -Fluorobenzenesulfonimide		
Experimental Protocols		

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